

~~X~~ Sum of 2 Vector Permutations
 n = length of vector / dimension / "sides on dice"

$$S_n = \left\{ \begin{array}{l} \text{all permutations of vectors in } \mathbb{F}^n \\ \text{using components } 1, 2, \dots, n \end{array} \right\}$$

$$|S_n| = n!$$

$$A(n) = \left| \left\{ \pi + \sigma : \pi, \sigma \in S_n \right\} \right|$$

t = length of a partial vector sum

$$A_t(n) = \left| \left\{ \pi + \sigma : \pi, \sigma \in S_n \text{ in } \mathbb{F}^t \right\} \right|$$

$$A_1(1),$$

$$A_1(2), A_2(2)$$

$$A_1(3), A_2(3), A_3(3)$$

$\vdots$

$$A_1(n) \cdot \cdot \cdot A_n(n)$$